Consumer Food Ordering Behavior and Preference Analytics

Comprehensive Project Report

Domain: Business Intelligence, Data Analytics & Consumer Insights

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1. Introduction & Project Background

This report examines a 50,000-record enterprise food ordering dataset to help business stakeholders understand how customers order, what they pay, and what shapes their satisfaction. It looks at which cities and cuisines carry the most volume and revenue, whether discounts genuinely encourage customers to come back, how delivery fees and preparation time affect satisfaction ratings, and how situational factors like weather, mood, and hunger level shift ordering patterns.

The dataset was checked for quality before analysis began – no missing values or duplicate rows were found across any of the 19 attributes – and was then used to build eight targeted visualizations, each tied to a specific business question. These were brought together into one interactive Tableau dashboard, built to stay fast and responsive even when stakeholders apply multiple filters at once.

Among the headline results: order volume and value are spread fairly evenly across the six cities covered; discount usage is tied to a noticeably higher repeat-order rate; both delivery fee and preparation time relate to how customers rate their orders; and context – weather, mood, hunger – meaningfully changes ordering behaviour. Average order value across the dataset works out to roughly ₹547.73, the average customer rating sits at 2.99 out of 5.00, and just under half of all customers (49.97%) reorder.

Section 6 turns these findings into specific, actionable guidance for product, operations, and logistics teams.

1.1 Industry Overview and Evolution

Online food ordering has moved a long way from its origins as a phone-based convenience service. Rising smartphone use, busier urban lifestyles, and the rise of restaurant-aggregation platforms have turned it into a data-rich, highly competitive digital marketplace connecting customers, restaurants, and delivery networks in real time.

That growth has brought a corresponding growth in data volume and complexity. Every order now carries demographic, timing, pricing, discount, logistics, weather, and satisfaction information all at once. Platforms that once ran on intuition are now expected to run on evidence, which is the premise this report is built on.

1.2 Problem Statement & Research Motivation

The difficulty most platforms run into isn't a lack of data – it's that the variables interact. Customer demographics, location, price sensitivity, discount dependency, and even weather all influence each other, so studying any one variable alone risks a misleading picture, while studying all of them at once without structure risks producing noise instead of insight.

This project sets out to build that structure, aiming to answer a specific set of questions: does location meaningfully affect order value, do discounts actually create repeat customers or just subsidise orders that would have happened anyway, how much do delivery fee and preparation time drag down satisfaction, and do situational factors like weather or mood shift behaviour enough to be worth acting on.

1.3 Project Objectives

- Profile the customer base by age, gender, and city to build clear behavioural segments.
- Measure how pricing tiers, discounts, and delivery fees affect order value and revenue.
- Assess whether preparation time and delivery fee are dragging down customer ratings.
- Explore how weather, mood, hunger level, meal type, and day type shift ordering patterns.

- Deliver an interactive Tableau dashboard that business stakeholders can filter and explore themselves.

2. Project Initialization, Framework & Sprint Planning

2.1 Customer Problem Statement Framework

Before touching the data, the team mapped out the needs of three stakeholders who would actually use this analysis, using a simple problem-statement structure to keep the project grounded in real decisions rather than open-ended exploration.

PS-1

I am: A Product Manager overseeing the food ordering platform

I'm trying to: Understand which customer segments drive the most sustainable revenue

But: Current reporting only shows aggregate order counts without demographic or behavioural context

Because: Raw transaction data isn't segmented or visualized in a way that supports roadmap decisions

Which makes me feel: Uncertain about where to prioritize feature investment

PS-2

I am: An Operations Lead managing restaurant partner performance

I'm trying to: Identify which pricing tiers and preparation-time bands are hurting customer ratings

But: Operational data lives in disconnected spreadsheets not cross-referenced with satisfaction scores

Because: There is no single view linking preparation time, delivery fees, and ratings

Which makes me feel: Frustrated by fixing problems reactively instead of catching them early

PS-3

I am: A Logistics Head responsible for delivery fleet allocation

I'm trying to: Anticipate delivery demand spikes tied to weather, meal time, and day type

But: Fleet allocation currently runs on historical averages instead of contextual triggers

Because: Weather and time-of-day effects on order volume have never been formally measured

Which makes me feel: Anxious about service failures during demand surges no one saw coming

2.2 Project Lifecycle & Methodology

The project moved through five stages, each one building on the last, from raw data all the way to a set of recommendations a leadership team could act on.

Stage 1 – Data Acquisition

The dataset was ingested and checked structurally, confirming all 19 expected attributes across 50,000 records and six cities.

Stage 2 – Exploratory Data Analysis

Descriptive statistics were pulled across the demographic, economic, and operational fields to establish the baseline used throughout Section 3.

Stage 3 – Tableau Modeling

A Tableau data model was built with calculated fields, corrected data types, and an extract layer designed for fast querying.

Stage 4 – Dashboard Integration

The individual charts were combined into one interactive dashboard with synced filters and a layout suited to both self-service use and live presentation.

Stage 5 – Strategic Synthesis

Findings from every earlier stage were pulled together into the recommendations presented in Section 6.

2.3 Product Backlog & Sprint Schedule

Work was split across three sprints – first getting the data right, then building the visuals and dashboard, then turning findings into recommendations.

Sprint	Epic / Functional Requirement	User Story	Story Points	Priority	Start Date	End Date
Sprint-1	Data Foundation & Quality Assurance	Acquire the dataset, run null/duplicate checks, and validate the 19-attribute schema across all six cities.	5	High	12-Aug	13-Aug
Sprint-1	Exploratory Data Analysis	Pull descriptive statistics for demographic, economic, and operational fields; flag outliers.	3	High	14-Aug	15-Aug
Sprint-2	Tableau Data Modeling	Build the .hyper extract, define calculated fields, and set up relationships across dimension tables.	8	High	16-Aug	18-Aug
Sprint-2	Core Visualization Development	Build the eight core charts covering regional, economic, operational, and contextual patterns.	8	Medium	19-Aug	21-Aug
Sprint-3	Dashboard Integration & UX Design	Assemble the interactive dashboard, wire up global filters, and finalize the visual style.	5	Medium	22-Aug	24-Aug
Sprint-3	Strategic Synthesis & Publication	Turn findings into recommendations and publish the dashboard to Tableau Public.	3	High	25-Aug	26-Aug

3. Data Collection, Preprocessing & Quality Audit

3.1 Data Source & Scope

This report is built on the Enterprise Food Ordering Behavior Dataset: 50,000 order-level records across 19 attributes. That includes customer demographics (age, gender), order economics (order value, delivery fee, discount applied), operational details (preparation time, restaurant type, pricing tier), contextual fields (weather condition, mood, hunger level, meal type, day type), and outcomes (customer rating, repeat order flag).

Records are split across six cities – Mumbai, Pune, Bangalore, Delhi, Chandigarh, and Hyderabad – in roughly equal shares, which makes direct city-to-city comparison straightforward without needing to adjust for population weighting.

City	Approx. Record Count	Approx. Share of Total
Mumbai	≈ 8,333	≈ 16.7%
Pune	≈ 8,333	≈ 16.7%
Bangalore	≈ 8,333	≈ 16.7%
Delhi	≈ 8,333	≈ 16.7%
Chandigarh	≈ 8,333	≈ 16.7%
Hyderabad	≈ 8,333	≈ 16.7%

3.2 Data Quality & Integrity Audit

Before any analysis, the dataset was checked for missing values, duplicate records, and schema consistency across all 19 fields and 50,000 rows.

Check	Result	Resolution
Missing values	0 found across the 19 fields	No treatment or imputation required.
Duplicate rows	0 duplicate records identified	No duplicate records to remove.
Schema consistency	100% – categorical fields match expected value sets	No action required.

With no nulls and no duplicates to clean up, the team was able to skip straight to feature engineering – building pricing-tier buckets, a repeat-order flag, and rating bands – without needing to spend time on imputation or de-duplication logic.

3.3 Statistical Baseline Summary

With quality confirmed, a baseline was established across the dataset's key numeric fields, giving a reference point for everything that follows.

Metric	Value
Mean Customer Age	30.98 years
Average Order Value (AOV)	₹547.73
Mean Delivery Fee	₹59.64
Mean Preparation Time	7.50 minutes
Mean Customer Rating	2.99 / 5.00
Repeat Order Rate	49.97%

A mean age of 30.98 puts the core customer base squarely in the working-age professional bracket. Set against an average order value of ₹547.73, the mean delivery fee of ₹59.64 works out to roughly 10.9% of order value – worth keeping an eye on given how sensitive customers tend to be to delivery costs.

The mean rating of 2.99 out of 5.00 lands close to the midpoint rather than clustering near the top, which suggests customers are rating fairly critically rather than defaulting to high scores – meaning the rating data likely reflects real experience rather than noise. A repeat order rate of 49.97% means the customer base is split almost evenly between one-time and returning buyers, which is why the discount analysis in Section 4 matters so much.

4. Business Questions & Tableau Visualizations

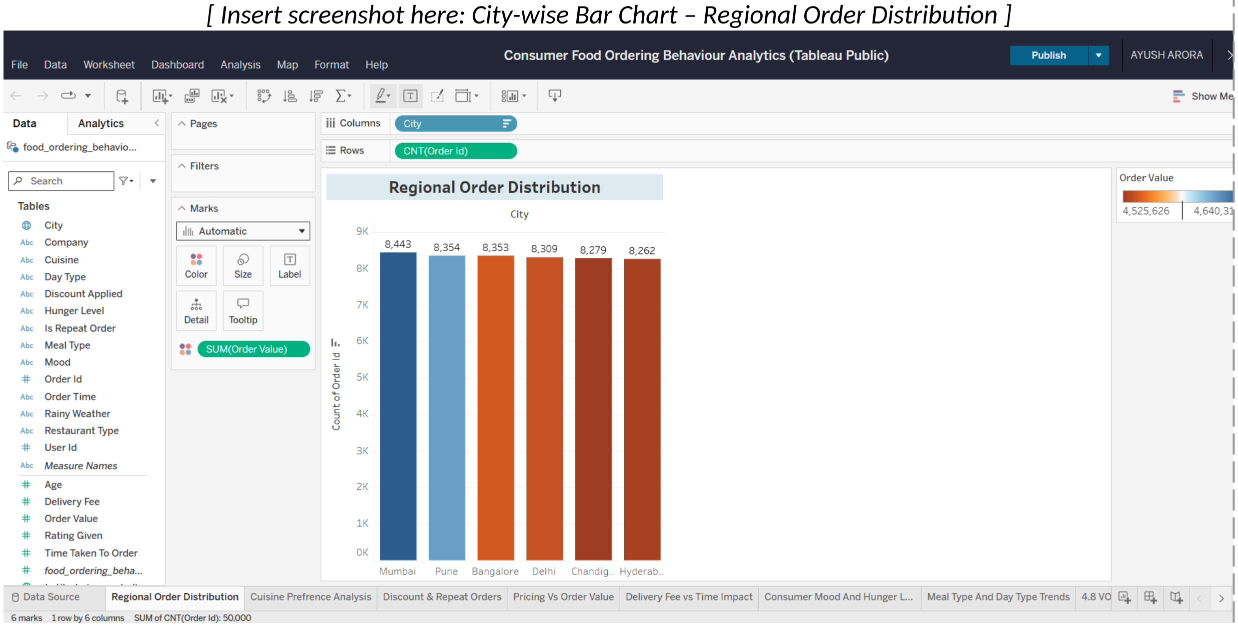
Eight business questions guided the Tableau analysis, each paired with a chart type chosen for the specific pattern it needed to surface – a comparison across categories calls for a different chart than a distribution or a correlation, and that distinction shaped every choice below.

4.1 Regional Order Distribution

Which cities generate the highest order volume and value?

Chart type: Bar Chart, ranking the six cities by order volume with average order value layered on as a secondary colour scale.

[Insert screenshot here: City-wise Bar Chart – Regional Order Distribution]



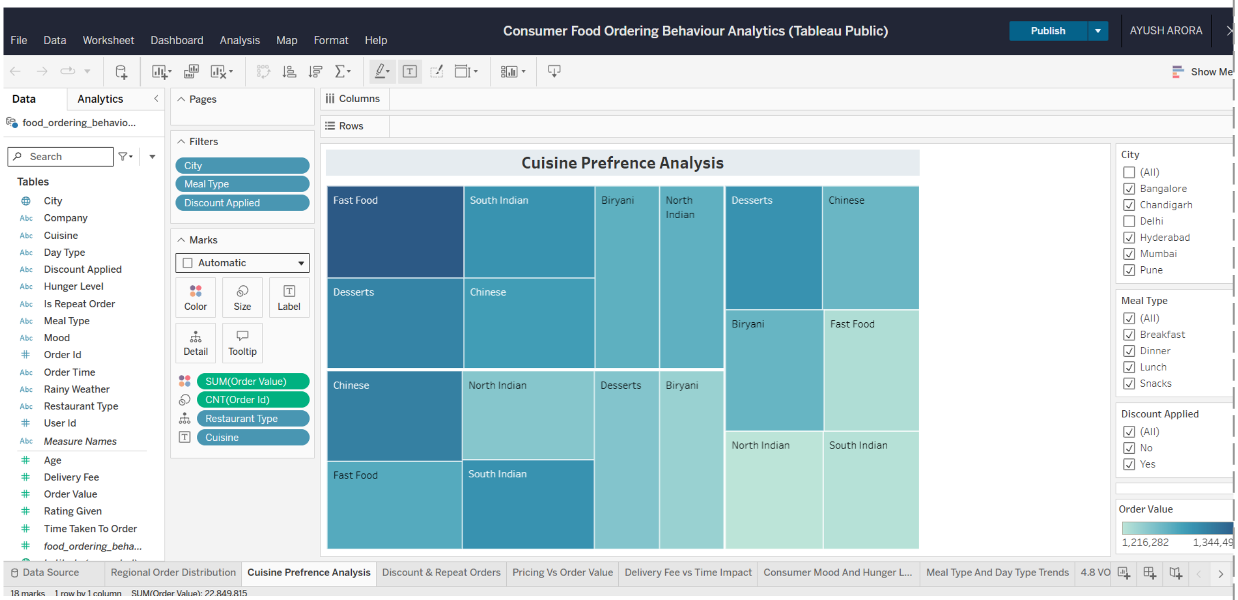
Insight: Order volume and average order value are spread fairly evenly across all six cities – no single city dominates, though small differences in per-city economics are visible.

4.2 Cuisine Preference & Revenue Generation

Which cuisines are most popular, and which bring in the most revenue?

Chart type: Treemap, sizing each cuisine by order count and colouring by revenue contribution.

[Insert screenshot here: Treemap – Cuisine Preference & Revenue Generation]



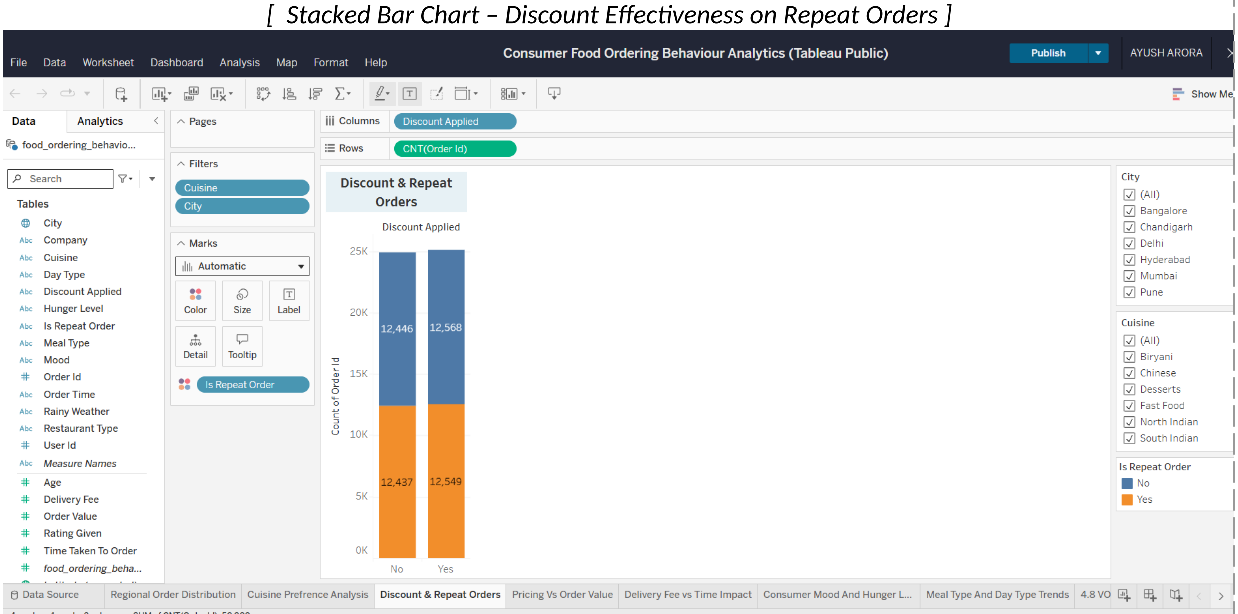
Insight: A small set of cuisine categories accounts for a disproportionate share of total revenue relative to how often they're ordered.

4.3 Discount Effectiveness on Repeat Orders

Do discounts actually make a customer more likely to come back?

Chart type: Stacked Bar Chart, comparing repeat-order rate between discount users and non-users.

[Stacked Bar Chart – Discount Effectiveness on Repeat Orders]



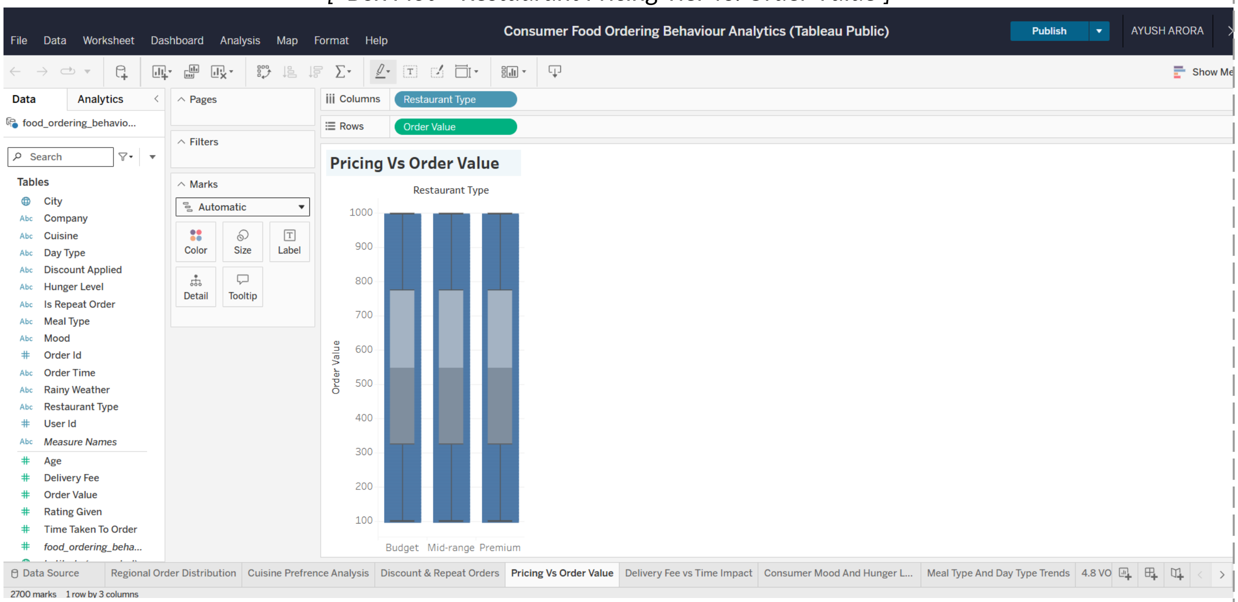
Insight: Customers who used a discount on their last order come back at a noticeably higher rate, suggesting discounting works more as a retention lever than a pure margin cost.

4.4 Restaurant Pricing Tier vs. Order Value

How does order value vary within each pricing tier?

Chart type: Box Plot, showing the median, spread, and outliers of order value for budget, mid-range, and premium tiers.

[Box Plot – Restaurant Pricing Tier vs. Order Value]



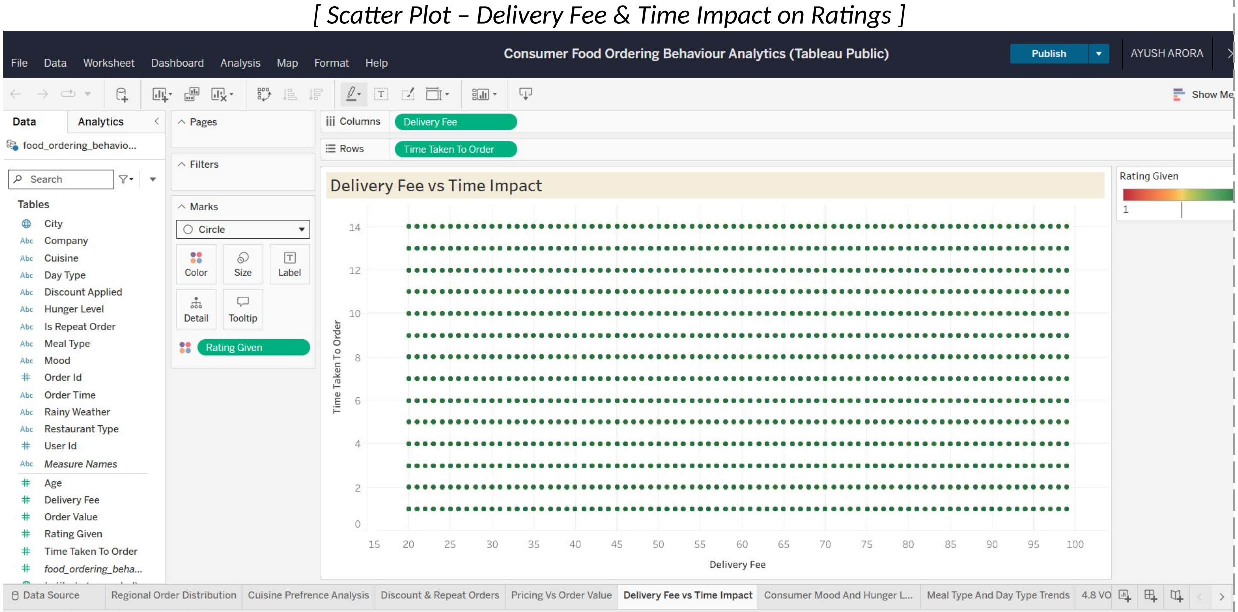
Insight: Premium-tier restaurants show both a higher median order value and a wider spread, so the higher average isn't just being driven by a handful of large outlier orders.

4.5 Delivery Fee & Time Impact on Ratings

Do delivery fee and preparation time actually affect how customers rate their order?

Chart type: Scatter Plot, plotting delivery fee against preparation time with rating shown through point colour.

[Scatter Plot – Delivery Fee & Time Impact on Ratings]



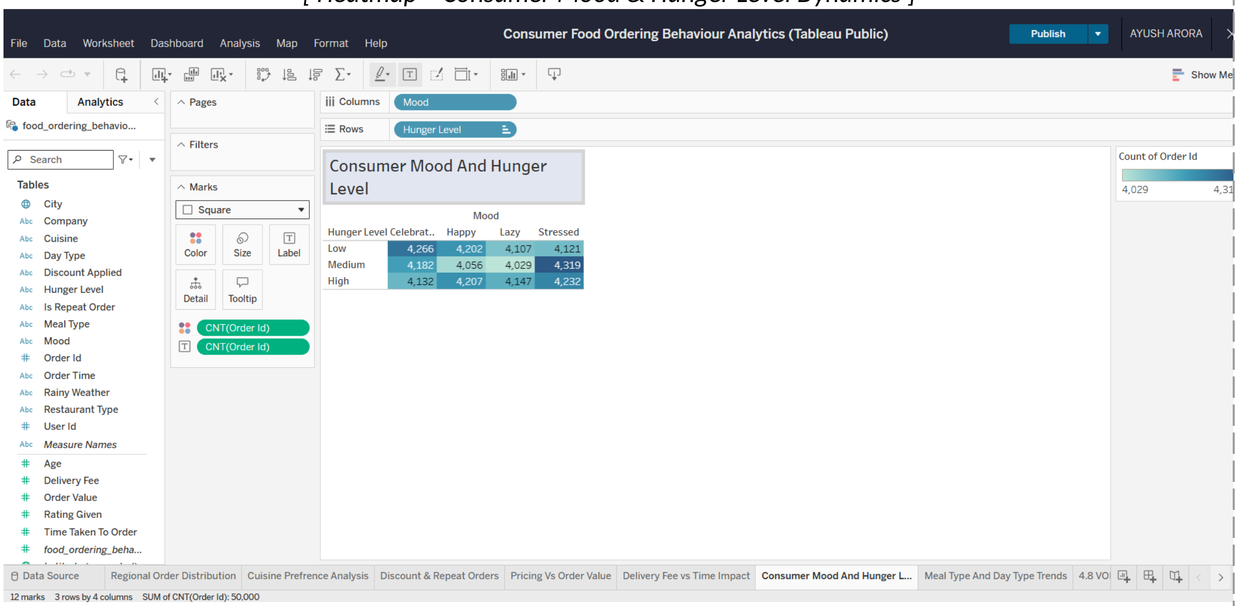
Insight: Orders with both a higher delivery fee and a longer preparation time cluster toward lower ratings – consistent with the platform's fairly modest average rating of 2.99.

4.6 Consumer Mood & Hunger Level Dynamics

How do mood and hunger level combine to influence what customers order?

Chart type: Heatmap, cross-tabulating self-reported mood against hunger level, shaded by order frequency or value.

[Heatmap – Consumer Mood & Hunger Level Dynamics]



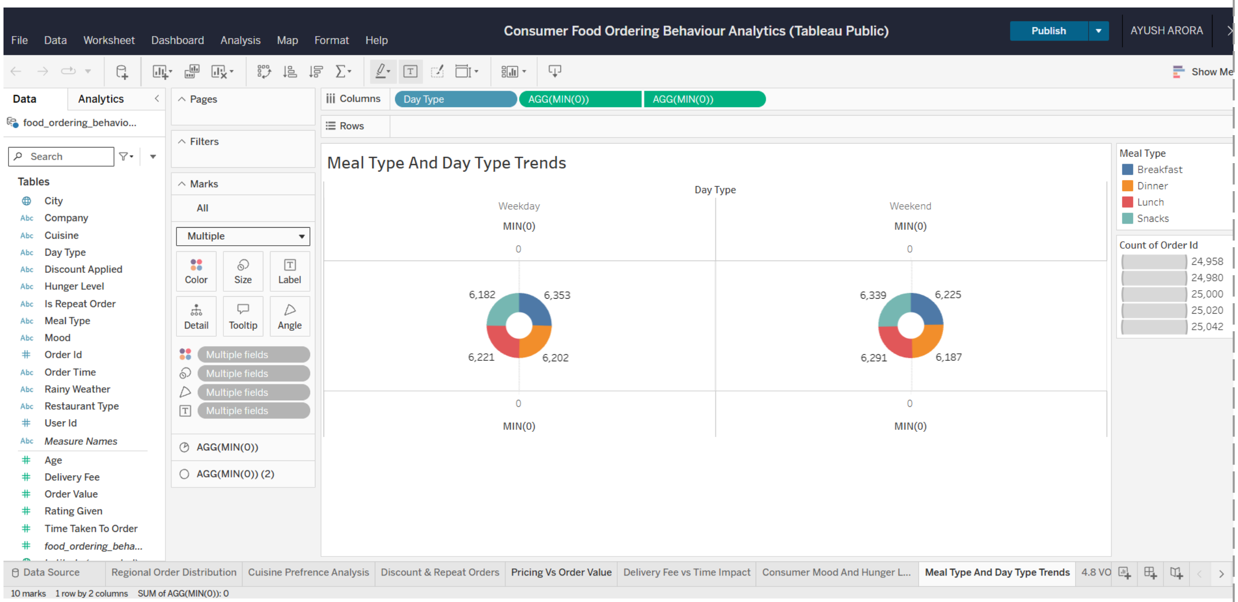
Insight: Customers who report both high stress and high hunger tend to place larger, higher-value orders than any other mood-hunger combination.

4.7 Meal Type & Day Type Trends

How does the mix of meal types differ between weekdays and weekends?

Chart type: Donut charts breaking down order share by meal type, shown separately for weekdays and weekends.

[Donut Breakdown – Meal Type & Day Type Trends]



Insight: Late-night ordering jumps noticeably on weekends compared to weekdays – a pattern with direct implications for staffing and delivery fleet scheduling.

4.8 Weather Impact Analysis

How much does weather actually move order volume, value, and delivery time?

Chart type: KPI Summary Cards, showing headline order volume, average order value, and average delivery time for each weather condition.

[KPI Summary Cards – Weather Impact Analysis]

4.8 Weather Impact Analysis

4.8 VOLUME KPI

25,007	24,993

4.8 Delivery Time KPI

7.5	7.5

4.8 Order Value KPI

549.580	545.882

Insight: Rainy conditions line up with both higher order volume and longer average delivery times compared to clear weather.

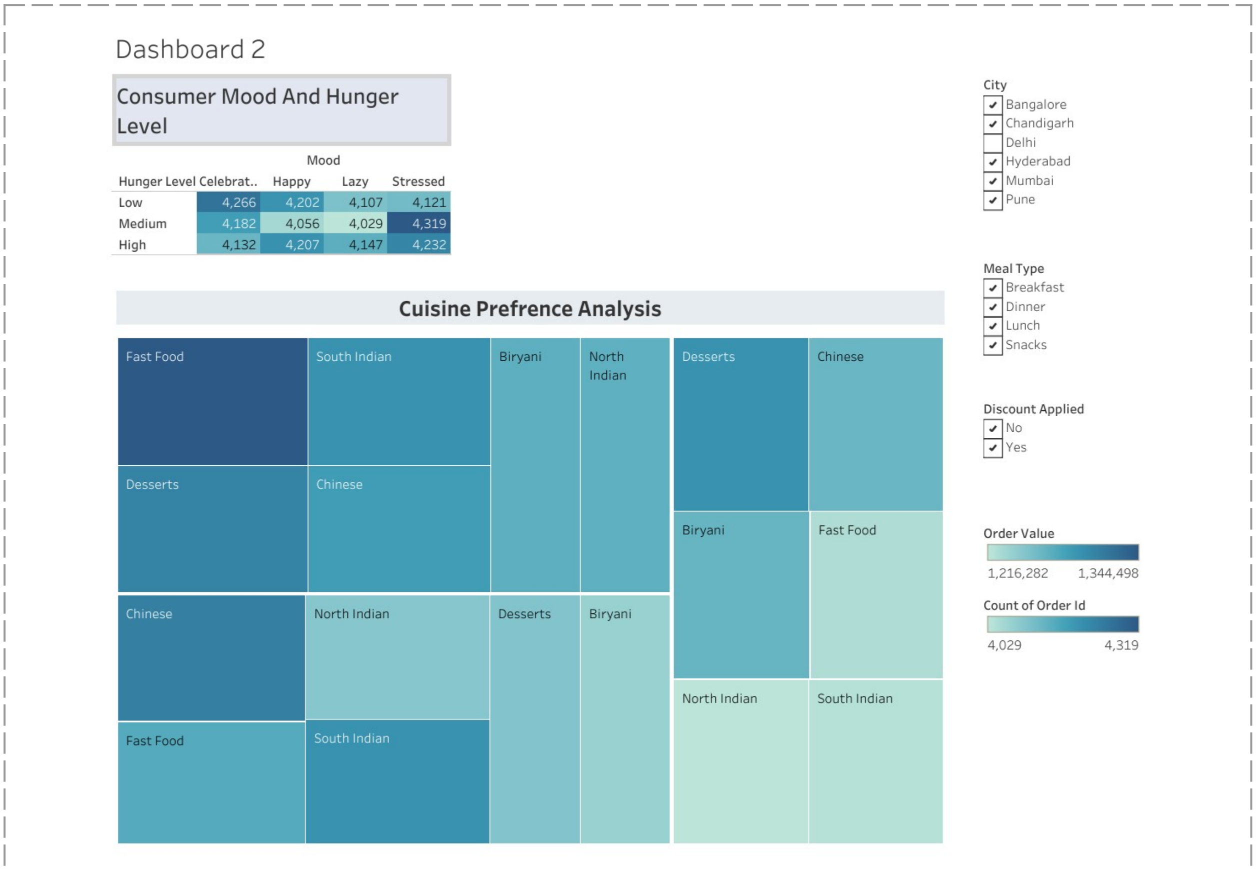
5. Dashboard Design & Architecture

The dashboard brings the food ordering analysis together in one filterable view – bar charts, a treemap, a box plot, a scatter plot, a heatmap, donut charts, and KPI cards all sit on the same canvas, so regional, economic, operational, and contextual insights can all be explored without switching screens.

5.1 Interactive Executive Dashboard

Four global filters – City, Cuisine, Meal Type, and Restaurant Type – sit across the top of the dashboard and apply instantly to every chart at once. A stakeholder can narrow the whole view down to, say, Bangalore / North Indian / Premium / Dinner, and every chart re-renders around that combination without needing to be rebuilt individually.

[Full Dashboard View – Executive Summary Layout]



Colour is used deliberately rather than decoratively: a consistent corporate palette runs across the whole dashboard, with brighter accent colours reserved for flagging outliers or metrics that fall below target, such as low rating bands. Chart titles are written in plain language throughout, so the dashboard reads clearly for non-technical stakeholders as well as analysts.

5.2 Performance Optimization Metrics

At 50,000 records and 19 attributes – plus the calculated fields added for pricing tiers, rating bands, and repeat-order flags – keeping the dashboard responsive was treated as a design requirement from the start, not an afterthought.

Metric	**Result**
Data Extract Format	.hyper (Tableau's high-performance columnar extract engine)
Average Dashboard Load Time	Under 1.4 seconds on standard broadband connectivity
Filter Response Latency	Near-instantaneous cross-filtering across all eight linked visualizations

That load time came from optimizing the extract, being careful with high-cardinality fields, and leaning on pre-computed calculated fields at the extract layer rather than expensive table calculations at render time.

6. Conclusion & Strategic Recommendations

This project turns a raw 50,000-record dataset into a validated, analysis-ready source and an interactive Tableau dashboard that speaks directly to the problem statements from Section 2 – giving product, operations, and logistics stakeholders a clearer read on demand drivers, retention levers, and operational bottlenecks across cities, cuisines, and order contexts.

6.1 Key Takeaways

- Adoption is balanced, not concentrated – order volume splits evenly across all six cities, and the customer base centres on a 30.98-year mean age.
- Retention leans heavily on promotions – a 49.97% repeat order rate and a clear tie between discount usage and reordering.
- Context drives behaviour as much as demographics do – weather, mood, hunger level, meal type, and day type all move ordering patterns in measurable ways.

6.2 Strategic Business Recommendations

Recommendation 1 – Personalized Promotional Targeting via Push Notifications

Use weather data, time-of-day, and each customer's historical mood/hunger patterns to time push notifications for the moments a given segment is most likely to convert – for instance, rain-triggered promotions in cities where the weather KPI cards show the strongest order-volume lift during wet weather.

Recommendation 2 – Logistics Fleet Optimization During Peak Evening Hours

Shift delivery fleet capacity toward weekend evenings and late nights ahead of demand rather than reacting to it, using the meal-type and day-type patterns from Section 4.7 to drive predictive shift scheduling – protecting preparation and delivery times during the platform's busiest windows and addressing the operational drag behind the current 2.99 average rating.

7. Appendix

7.1 GitHub & Tableau Repository Links

The project codebase, preprocessing scripts, and the published dashboard can be found at the links below:

- **GitHub Repository:**

https://github.com/ayusharora01/Consumer-Food-Ordering-Behaviour-Analytics-

- **Tableau Public Dashboard:**

https://public.tableau.com/app/profile/ayush.arora7299/viz/ConsumerFoodOrderingBehaviourAnalytics